JI-HOON KIM

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EDUCATION:

09/2005 - 06/2011	Ph. D., Department of Physics, Stanford University
03/1998 - 02/2002	B. Sc., School of Physics, *summa cum laude*, Seoul National University (SNU)

PROFESSIONAL APPOINTMENTS:

09/2026 -	Associate Chair, Department of Physics & Astronomy, Seoul National University
03/2024 -	Team Leader, Center for Data-driven Research for Materials and Universe, Institute for Data Innovation in Science, Seoul National University
03/2022 -	**Associate Professor,** Seoul National University
03/2018 - 02/2022	Assistant Professor, Seoul National University
07/2016 - 11/2017	**Research Associate,** Stanford University/KIPAC
07/2015 - 06/2017	**Einstein Fellow,** Stanford University/SLAC National Accelerator Laboratory
07/2014 - 06/2015	Einstein Fellow, California Institute of Technology
01/2014 - 06/2014	**Moore Fellow,** California Institute of Technology
11/2013 - 12/2013	Visiting Scholar, Stanford University/KIPAC
09/2011 - 10/2013	**IMPS Postdoctoral Fellow,** University of California at Santa Cruz
09/2005 - 08/2011	Research Assistant, Stanford University

AWARDS AND FELLOWSHIPS (SELECTED):

03/2022	College of Natural Sciences Research Award, Seoul National University
03/2021	College of Natural Sciences Education Award, Seoul National University
01/2020 - 12/2022	POSCO Science Fellowship, POSCO TJ Park Foundation
09/2019	College of Natural Sciences Best Lecture Award, Seoul National University
12/2018 - 11/2023	Principal Investigator, Samsung Science & Technology Foundation
07/2014 - 06/2017	Einstein Postdoctoral Fellowship, NASA
01/2014 - 06/2014	Moore Postdoctoral Fellowship, California Institute of Technology
09/2005 - 03/2009	William R. and Sara Hart Kimball Graduate Fellowship, Stanford University
03/2000 - 02/2002	Korea Foundation for Advanced Studies Fellowship

RESEARCH EXPERIENCES:

2022 - **Associate Professor**, Dept. of Physics and Astronomy, Seoul National University

- ***AGORA Project Coordinator,*** *Leading an inter-institutional collaboration for high-resolution simulations comparison since 2012 (160+ participants from 60+ institutions), having led or co-led 9+ papers, helped organize 16 workshops with 30+ attendees, and 150+ online conferences*
- ***PI, Samsung Science & Technology Foundation*** *"Towards the Multi-scale Understanding*

of the Growth of Supermassive Black Holes" (12/2018-11/2023)

- ***PI, KISTI KSC*** *"Inferring Cosmological Parameters on Large-scale Structures via Domain-Generalized Neural Networks and Lightcone Simulations"*
  (21,760,000 SUs on Nurion@KSC, Neuron@KSC, 20,000.0 GB on Nurion@KSC, 09/2024-08/2025)
- ***PI, KISTI KSC*** *"Multi-scale Numerical Experiment on Growth of SMBHs In Nuclear SCs"*
  (4,352,000 SUs on Nurion@KSC, Neuron@KSC, 20,000.0 GB on Nurion@KSC, 01/2023-12/2023)
- ***PI, KISTI KSC*** *"Multi-scale Numerical Experiment on Growth of SMBHs Enhanced by TDEs"*
  (6,528,000 SUs on Nurion@KSC, 20,000.0 GB on Nurion@KSC, 01/2022-12/2022)
- *Studying the co-evolution of massive black holes, nuclear star clusters, and their host galaxies*

2018 - 2022 **Assistant Professor**, Dept. of Physics and Astronomy, Seoul National University

- ***PI, KISTI KSC*** *"Multi-scale Numerical Experiment on Growth of SMBHs and Their Environment"*
  (7,834,000 SUs on Nurion@KSC, 20,000.0 GB on Nurion@KSC, 01/2021-12/2021)
- ***PI, KISTI KSC*** *"Multi-scale Numerical Experiment of Co-evolution of SMBHs and Their Hosts"*
  (6,528,000 SUs on Nurion@KSC, 15,000.0 GB on Nurion@KSC, 01/2020-12/2020)
- ***PI, KISTI KSC*** *"Towards the Multi-scale Numerical Understanding of SMBHs at Galaxy Centers"*
  (4,787,000 SUs on Nurion@KSC, 15,000.0 GB on Nurion@KSC, 12/2018-11/2019)
- ***PI, KISTI KSC*** *"Towards the Understanding of Growth & Evolution of SMBHs at Galaxy Centers"*
  (400,000 SUs on Tachyon-II@KSC, 2,000.0 GB on Tachyon-II@KSC, 07/2018-10/2018)
- *Simulating high-redshift quasar hosts with massive black holes and star-forming molecular clouds*

2015 - 2017 **Research Associate / Postdoctoral Researcher**, Stanford University/SLAC (Mentor: T. Abel)

- ***Co-I, NASA HEC*** *"FIRE: DM and Galaxy Formation with Unprecedented Physics and Resolution"*
  (3,166,000 SUs on Pleiades@NAS, 100,000.0 GB on Pleiades@NAS, 11/2016-09/2019)
- ***PI, NSF XSEDE*** *"Resolving the Impact of Supermassive Black Hole & Stellar Physics on Galaxies"*
  (3,234,000 SUs on Stampede@TACC, 50,000.0 GB on Ranch@TACC, 10/2015-09/2016)
- ***Co-I, NSF XSEDE*** *"Simulating the Local Group"*
  (3,649,000 SUs on Stampede@TACC, 50,000.0 GB on Ranch@TACC, 10/2016-09/2017)
- *Modeling the accretion and feedback of massive black hole seeds in the high-redshift universe*

2014 - 2015 **Postdoctoral Researcher**, California Institute of Technology (Mentor: P. Hopkins)

- ***PI, NSF XSEDE*** *"Resolving the Impact of Supermassive Black Holes on Galaxies"*
  (1,200,000 SUs on Stampede@TACC, 20,000.0 GB on Ranch@TACC, 10/2014-09/2015)
- *Simulating and analyzing the formation of star clusters in high-redshift proto-galaxies*

2013 Visiting Scholar, KIPAC/Stanford University

2011 - 2013 **Postdoctoral Researcher**, UC Santa Cruz (Mentors: M. Krumholz & J. Primack)

- ***Co-I, NSF XSEDE*** *"Star Formation in Galaxies: From Recipes to Real Physics"*
  (on Stampede/Ranger@TACC, Pleiades@NASA, Kraken@NICS)
- *Modeling radiative feedback of star-forming molecular clouds in galaxy formation*

2006 - 2011 **Research Assistant**, KIPAC/SLAC/Stanford University (Advisor: T. Abel)

- *Modeling the accretion and feedback of massive black holes, formation and feedback of stars*

2006 Research Assistant, KIPAC/Stanford University (Advisor: S. Church)

• *Building data pipelines for the QUAD Collaboration*

2005 Research Assistant, KIPAC/Stanford University (Advisor: E. Bloom)

• *Studying the cross-correlation between GRBs and SNe Ic events for the Fermi/GLAST Collaboration*

PEER-REVIEWED PUBLICATIONS [* = corresponding author] :

• **45)** Lee, J. -Y., **Kim, J. -H.*,** Kim, G., Ting, Y. -S., Chang, W., & Sohn, J., "Neural Point-Cloud Summaries Beyond the Power Spectrum: Simulation-Based Inference from SDSS BOSS LOWZ", *ApJ submitted (2026)*

• **44)** Lee, J. et al. including **Kim, J. -H.**, "The DARWIN Project: Cosmological Simulations of Galaxy Formation down to Dwarf Scales in Diverse Environments", *astro-ph:2610.25786, ApJ submitted (2026)*

• **43)** Jung, M. et al. including **Kim, J. -H.**, "FOGGIE XVI: Ram-Pressure Stripping and Absorption Signatures of Dwarf-Satellite Gas in Milky Way-like Halos", *ApJ submitted (2026)*

• **42)** Kim, Y., Hwang, H. S., Hong, S. E., Jeong, D., Yang, M., & **Kim, J. -H.**, "Three-dimensional Dark Matter Mapping in the Solar Neighborhood with Gaia Data Release 3 and Deep Learning", *MNRAS submitted (2026)*

• **41)** Barrow, K., Nguyen, T. H., Roca-Fabrega, S., **Kim, J. -H.**, & 27 other co-authors for the *AGORA* Collaboration, "The AGORA High-resolution Galaxy Simulations Comparison Project. XI: Solving the Non-Spherical Morphology and Evolution of Dark Matter Halos with Haskap Pie", *astro-ph:2605.24097, ApJ submitted (2026)*

• **40)** Kim, D., Jo, Y., Hwang, H. S., **Kim, J. -H.**, Kim, J., & 24 other co-authors, "Constructing a Mock Galaxy Catalog for the All-Sky Spectroscopic Survey (A-SPEC) Using Machine-assisted Semi-Simulation Model", *astro-ph:2607.06933, ApJS 285 (2026) 46*

• **39)** Chung, E., Jo, Y., **Kim, J. -H.***, Jung, M., & Kwon, O. -K., "From Dense Gas Clouds to Supermassive Black Hole Seeds: Self-consistent Simulations of High-Redshift Star Cluster Evolution", *astro-ph:2606.10000, ApJ 1004 (2026) 250*

• **38)** Jo, Y., Jung, M., Bryan, G. L., Kim, S., **Kim, J. -H.**, & Lee, A., "Evolution of Nuclear Star Cluster in Dwarf Galaxy through Mergers and In-Situ Star Formation", *astro-ph:2603.18242, ApJ 1005 (2026) 103*

• **37)** Kim, H., **Kim, J. -H.***, Jung, M., Roca-Fabrega, S., & 21 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. X: Formation and Evolution of Galaxies at the High-redshift Frontier", *astro-ph:2511.04435, ApJ 1000 (2026) 276*

• **36)** Nguyen, T. H., Barrow, K., Jung, M., Rodriguez-Cardoso, R., Roca-Fabrega, S., **Kim, J. -H.**, & 19 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. IX - Part 2: Effects of a Major Galaxy Merger on the Stellar and Gas Morphology of a Milky Way-mass Galaxy Progenitor", *astro-ph:2607.21710, MNRAS submitted (2026)*

• **35)** Nguyen, T. H., Barrow, K., Jung, M., Rodriguez-Cardoso, R., Roca-Fabrega, S., **Kim, J. -H.**, & 19 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. IX - Part 1: Effects of a Major Galaxy Merger on Star Formation of a Milky Way-mass Galaxy Progenitor", *astro-ph:2607.21709, MNRAS submitted (2026)*

• **34)** Jung, M., **Kim, J. -H.***, Nguyen, T. H., Rodriguez-Cardoso, R., Roca-Fabrega, S., & 22 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. VIII: Disk Formation and Evolution of Simulated Milky Way Mass Galaxy Progenitors at $1 < z < 5$", *astro-ph:2505.05720, ApJ 994 (2025) 245*

• **33)** Rodriguez-Cardoso, R., Roca-Fabrega, S., Jung, M., Nguyen, T. H., **Kim, J. -H.**, & 17 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. VII:

Satellite Quenching in Zoom-in Simulation of A Milky Way-mass Halo", *astro-ph:2505.05844, A&A 698 A303 (2025)*

● **32)** Lee, S., Lee, H. M., **Kim, J. -H.***, Spurzem, R., Hong, J., & Chung, E., "Formation and Evolution of Compact Binaries Containing Intermediate Mass Black Holes in Dense Star Clusters", *astro-ph:2503.22109, ApJ 988 (2025) 15*

● **31)** Lee, J. -Y., **Kim, J. -H.***, Jung, M., & 6 other co-authors, "Inferring Cosmological Parameters on SDSS via Domain-Generalized Neural Networks and Lightcone Simulations", *astro-ph:2409.02256, ApJ 975 (2024) 38*

● **30)** Jo, Y., Kim, S., **Kim, J. -H.**, & Bryan, G. L., "Evolution of Star Clusters Within Galaxies Using Self-consistent Hybrid Hydro/*N*-body Simulation", *astro-ph:2408.03128, ApJ 974 (2024) 193*

● **29)** Lee, J., Shin, E. -J., **Kim, J. -H.**, Shapiro, P. R., & Chung, E., "Multiple Beads-on-a-string: Dark Matter-Deficient Galaxy Formation In A Mini-bullet Satellite-satellite Galaxy Collision", *astro-ph:2312.11350, ApJ 966 (2024) 72*

● **28)** Jung, M., **Kim, J. -H.***, Oh, B. K., Hong, S. E., Lee, J., & Kim, J., "Merger-tree-based Galaxy Matching: A Comparative Study Across Different Resolutions", *astro-ph:2312.02466, ApJ 965 (2024) 156*

● **27)** Strawn, C., Roca-Fabrega, S., Primack, J. R., **Kim, J. -H.***, & 24 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. VI: Similarities and Differences in the Circumgalactic Medium", *astro-ph:2402.05246, ApJ 962 (2024) 29*

● **26)** Jung, M., Roca-Fabrega, S., **Kim, J. -H.***, & 18 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo", *astro-ph:2402.05392, ApJ 964 (2024) 123*

● **25)** Roca-Fabrega, S., **Kim, J. -H.***, Primack, J. R., Jung, M., & 23 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. IV: Halo and Galaxy Mass Assembly in a Cosmological Zoom-in Simulation at $z \leq 2$", *astro-ph:2402.06202, ApJ 968 (2024) 125*

● **24)** Jo, Y. et al. including **Kim, J. -H.**, "Calibrating Cosmological Simulations with Implicit Likelihood Inference Using Galaxy Growth Observables", *astro-ph:2211.16461, ApJ 944 (2023) 67*

● **23)** Lee, S., **Kim, J. -H.***, & Oh, B. K., "Growth of A Massive Black Hole In A Dense Star Cluster Via Tidal Disruption Accretion", *astro-ph:2211.02376, ApJ 943 (2023) 77*

● **22)** Shin, E. -J., Tacchella, S., **Kim, J. -H.**, Iyer, K. G., & Semenov, V. A., "Star Formation Variability As A Probe For the Baryon Cycle Within Galaxies", *astro-ph:2211.01922, ApJ 947 (2023) 61*

● **21)** Oh, B. K., An, H., Shin, E. -J., **Kim, J. -H.***, & Hong, S. E., "Machine-guided Exploration and Calibration of Astrophysical Simulations", *astro-ph:2203.06914, MNRAS 515 (2022) 693*

● **20)** Lee, J., Shin, E. -J., & **Kim, J. -H.***, "Dark Matter Deficient Galaxies And Their Member Star Clusters Form Simultaneously During High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations", *astro-ph:2108.01102, ApJ Letters 917 (2021) L15*

● **19)** Roca-Fabrega, S., **Kim, J. -H.***, Hausammann, L., Nagamine, K., & 19 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. III: Cosmological Zoom-in Simulation of A Milky Way-mass Halo", *astro-ph:2106.09738, ApJ 917 (2021) 64*

● **18)** Shin, E. -J., **Kim, J. -H.***, & Oh, B. K., "How Metals Are Transported In And Out Of A Galactic Disk: Dependence On The Hydrodynamic Schemes In Numerical Simulations", *astro-ph:2106.04640, ApJ 917 (2021) 12*

● **17)** Shin, E. -J., Jung, M., Kwon, G., **Kim, J. -H.***, Lee, J., Jo, Y., & Oh, B. K., "Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions in High-resolution Numerical Simulations", *astro-ph:2007.09889, ApJ 899 (2020) 25*

• **16)** Ma, X. et al. including **Kim, J. -H.**, "Self-consistent Proto-Globular Cluster Formation in Cosmological Simulations of High-redshift Galaxies", *astro-ph:1906.11261, MNRAS 493 (2020) 4315*

• **15) Kim, J. -H.***, Wise, J. H., Abel, T., Jo, Y., Primack, J. R., & Hopkins, P. F., "High-redshift Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Stellar Feedback and Quasar Growth", *astro-ph:1910.12888, ApJ 887 (2019) 120*

• **14)** Jo, Y., & **Kim, J. -H.***, "Machine-assisted Semi-Simulation Model (MSSM): Estimating Galactic Baryonic Properties from Their Dark Matter Using A Machine Trained on Hydrodynamic Simulations", *astro-ph:1908.09844, MNRAS 489 (2019) 3565*

• **13)** Bryan, G. L. et al. including **Kim, J. -H.** for the *ENZO* Collaboration, "*Enzo*: An Adaptive Mesh Refinement Code for Astrophysics (Version 2.6)", *JOSS 4(42) (2019) 1636*

• **12) Kim, J. -H.***, Ma, X., Grudic, M. Y., Hopkins, P. F., Hayward, C. C., & 5 other co-authors for the *FIRE* Collaboration, "Formation of Globular Cluster Candidates in Merging Proto-galaxies at High Redshift: A View from the *FIRE* Cosmological Simulations", *astro-ph:1704.02988, MNRAS 474 (2018) 4232*

• **11)** Hopkins, P. F. et al. including **Kim, J. -H.**, "The *FIRE*-2 Simulations: Physics Versus Numerics in Galaxy Formation", *astro-ph:1702.06148, MNRAS 480 (2018) 800*

• **10)** Butsky, I., Zrake, J., **Kim, J. -H.**, Yang, H. -I., & Abel, T., "Ab Initio Simulations of A Supernova Driven Galactic Dynamo in An Isolated Galaxy", *astro-ph:1610.08528, ApJ 843 (2017) 113*

• **9) Kim, J. -H.***, Agertz, O., Teyssier, R., Butler, M. J., Ceverino, D., & 38 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project. II: Isolated Disk Test", *astro-ph:1610.03066, ApJ 833 (2016) 202*

• **8)** Wetzel, A., Hopkins, P. F., **Kim, J. -H.**, Faucher-Giguere, C-A., Keres, D., & Quataert, E., "Reconciling Dwarf Galaxies with LCDM Cosmology: Simulating A Realistic Population of Satellites Around A Milky Way-Mass Galaxies", *astro-ph:1602.05957, ApJ Letters 827 (2016) L23*

• **7) Kim, J. -H.***, Abel, T., Agertz, O., Bryan, G. L., Ceverino, D., & 41 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project", *astro-ph:1308.2669, ApJS 210 (2014) 14*

• **6)** Bryan, G. L. et al. including **Kim, J. -H.** for the *ENZO* Collaboration, "*Enzo*: An Adaptive Mesh Refinement Code for Astrophysics", *astro-ph:1307.2265, ApJS 211 (2014) 19*

• **5) Kim, J. -H.***, & Lee, J., "How Does the Surface Density and Size of Disk Galaxies Measured in Hydrodynamic Simulations Correlate with the Halo Spin Parameter?", *astro-ph:1210.8321, MNRAS 432 (2013) 1701*

• **4) Kim, J. -H.***, Krumholz, M. R., Wise, J. H., Turk, M. J., Goldbaum, N. J., & Abel, T., "Dwarf Galaxies with Ionizing Radiation Feedback. II: Spatially-resolved Star Formation Relation", *astro-ph:1210.6988, ApJ 779 (2013) 8*

• **3) Kim, J. -H.***, Krumholz, M. R., Wise, J. H., Turk, M. J., Goldbaum, N. J., & Abel, T., "Dwarf Galaxies with Ionizing Radiation Feedback. I: Escape of Ionizing Photons", *astro-ph:1210.3361, ApJ 775 (2013) 109*

• **2) Kim, J. -H.***, Wise, J. H., Alvarez, M. A., & Abel, T., "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes. I: Feedback-regulated Star Formation and Black Hole Growth", *astro-ph:1106.4007, ApJ 738 (2011) 54*

• **1) Kim, J. -H.***, Wise, J. H., & Abel, T., "Galaxy Mergers with Adaptive Mesh Refinement: Star Formation and Hot Gas Outflow", *astro-ph:0902.3001, ApJ Letters 694 (2009) L123*

OTHER CONTRIBUTIONS:

● **7)** Roca-Fabrega, S., **Kim, J. -H.**, Primack, J. R., & 9 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project: CosmoRun Public Data Release", *astro-ph:2408.00432*

● **6)** Shin, E. -J., & **Kim, J. -H.***, "News and Views: Giant Collision Created Galaxies Devoid of Dark Matter", *Nature 605 (2022) 427*

● **5)** Roca-Fabrega, S., **Kim, J. -H.**, Primack, J. R., & 11 other co-authors for the *AGORA* Collaboration, "The *AGORA* High-resolution Galaxy Simulations Comparison Project: Public Data Release", *astro-ph:2001.04354*

● **4)** Pineda, J. L. et al. including **Kim, J. -H.**, "Bridging the Gap: Observations and Theory of Star Formation Meet on Large and Small Scales", *Keck Institute for Space Studies Report, Pasadena, CA, November 2014*

● **3)** Kaehler, R., Abel, T., & **Kim, J. -H.**, "Visualization of a High-resolution Galaxy Formation Simulation", *SuperComputing '11 Scientific Visualization Companion Proceedings, pp. 133-134, Seattle, WA, November 2011*

● **2) Kim, J. -H.**, "Galaxy Formation and Mergers with Stars and Massive Black Holes", *Doctoral Dissertation, Stanford University, Stanford, CA, May 2011*

● **1) Kim, J. -H.**, Wise, J. H., & Abel, T., "Galaxy Evolution on Adaptive Mesh Refinement", *First Stars III Conference, AIP Conference Proceedings, Vol. 990 (2008) pp. 429-431, Santa Fe, NM, July 2007*

PROFESSIONAL AND OUTREACH SERVICES:

2025	Organizer, 4th SNUARC Woojeon Astronomy Workshop, SNU (Aug. 21-22, 2025)
2025	**Organizer,** 13th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 8-9, 2025)
2025	Member, Scientific Organizing Committee for "International Workshop on Galaxy Formation + AGORA in Asia 2025", Osaka, Japan (May 26-30, 2025)
2024	Organizer, 12th Workshop for the *AGORA* Project, online (Aug. 8, 2024)
2024	**Organizer,** 2nd CTP Bosan Workshop: AGORA in Asia + 5th Numerical Galaxy Formation Meeting, SNU (May 7-10, 2024)
2024 - 2027	Professional Committees Served: Academic Affairs/Korean Astronomical Society
2023	Member, Scientific Organizing Committee for "Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback", Hangzhou, China (Oct. 10-13, 2023)
2023	Organizer, 11th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 11-12, 2023)
2023 - 2025	Organizer, Kyewon Physics Public Lecture, Center for Theoretical Physics, SNU
2023	Co-organizer, 4th Numerical Galaxy Formation Workshop at Konjiam (Jan. 11, 2023)
2023 - 2026	**Member, Board of Directors,** Korean Astronomical Society
2022	Organizer, 10th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 19-20, 2022)
2022	Mentor, SNU Science Education Institute for The Gifted (Physics)
2022	Organizer, 8th Korea Astronomy Machine Learning Meeting, online (Feb. 22, 2022)
2022	Speaker, SNU-Naver Lecture Series: One Thousand Keywords (Feb. 11, 2022)
2022	Organizer, 3rd Numerical Galaxy Formation Workshop, online (Jan. 25, 2022)
2021	Organizer, 9th Workshop for the *AGORA* Project, online (Aug. 12-13, 2021)
2021	Speaker, KAOS Foundation Science Concert (*Knowledge Awakening On Stage;* Jul. 28, 2021)
2021	Co-organizer, 2nd Numerical Galaxy Formation Workshop, online (Jan. 28, 2021)
2020	Organizer, 1st Numerical Galaxy Formation Workshop, SNU (Jan. 16, 2020)
2019	Organizer, 8th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 9-10, 2019)

2019 - 2022	Member, National Organizing Committee for the International Astronomical Union General Assembly 2022, Busan, Korea (Aug. 2-11, 2022)
2018 -	Member, Korea e-Science Forum, Korea Institute of Science and Technology Information
2018	Organizer, 7th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 10-11, 2018)
2018 -	Department Committees Served: Graduate Admission and Qualification, Academic Affairs, Student Affairs and Scholarship, Department Planning, Faculty Search and Hiring
2018 -	**Professional Memberships**: International Astronomical Union, Korean Physical Society, Korean Astronomical Society
2017	Organizer, 6th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 11-13, 2017)
2016	Expert Referee, DiRAC Resource Allocation Committee, the United Kingdom
2016	External Review Panel, NASA Postdoctoral Program Fellowship
2016	Organizer, 5th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 12-14, 2016)
2015	Organizer, 4th Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 21-23, 2015)
2015	External Review Panel, Department of Energy ASCR Leadership Computing Challenge
2015	External Review Panel, NASA Earth and Space Science Fellowship
2014	Organizer, 3rd Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 15-17, 2014)
2014 -	**Referees Served**: Astrophysical Journal, Astrophysical Journal Letters, Monthly Notices of the Royal Astronomical Society, Astronomy and Astrophysics, Astronomy and Computing, Open Journal of Astrophysics, Journal of the Korean Astronomical Society
2013	Organizer, 2nd Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 16-18, 2013)
2012	External Review Panel, NASA Earth and Space Science Fellowship
2012	Organizer, 1st Workshop for the *AGORA* Project at UC Santa Cruz (Aug. 17-19, 2012)
2012 -	**Project Coordinator,** *AGORA* High-resolution Galaxy Simulations Comparison Project
2010	Provided a movie data of cosmological galaxy mergers and structure formation for the SLAC booth in the SuperComputing '10 conference (SC2010), New Orleans, LA
2010	Provided a movie data of cosmological galaxy mergers for the Hayden Planetarium at the American Museum of Natural History, NY in the program of "The Big Bang"
2007 - 2011	KIPAC/SLAC Visualization Laboratory Presenter, Laboratory Tour Guide
2007 - 2009	Volunteer, SLAC Kids' Day
2002 - 2004	Lieutenant, Company Commander & Battalion Staff Officer, the Republic of Korea Army

TEACHING EXPERIENCES:

2026	**Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*
2026	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2026, *Artificial Intelligence in Astronomy*
2026	**Lecturer,** 1 semester, Seoul National University, *Classical Mechanics I*
2026	Lecturer, Korea Physics Olympiad (KPhO) Winter School 2026, *Mechanics and Astrophysics*
2025	**Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*

2025	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2025, *Numerical Experiments in Astrophysics and Cosmology*
2025	Lecturer, Korea Physics Olympiad (KPhO) Winter School 2025, *Mechanics and Astrophysics*
2024	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2024, *Reproducibility in Astrophysical Simulations*
2024	**Lecturer,** 2 semesters, Seoul National University, *Physics I & II (for physics/astronomy majors)*
2023	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2023, *Back of the Envelope (Astro)Physics*
2023	**Lecturer,** 2 semesters, Seoul National University, *Physics I & II (for physics/astronomy majors)*
2022	**Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*
2022	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2022, *Supermassive Black Holes*
2022	**Lecturer,** 1 semester, Seoul National University, *Physics I*
2021	**Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*
2020	**Lecturer,** 2 semesters, Seoul National University, *Classical Mechanics I & II*
2019	Lecturer, Korea Institute for Advanced Studies - SNU Physics Winter Camp 2019, *Astrophysical Black Holes*
2019	**Lecturer,** 2 semesters, Seoul National University, *Classical Mechanics I & II*
2018	Lecturer, Korea Institute for Advanced Studies - SNU Physics Winter Camp 2018, *Astrophysical Dark Matter*
2018	**Lecturer,** 1 semester, Seoul National University, *Topics in Modern Astrophysics and Cosmology*
2018	Guest Lecturer, Seoul National University, *Classical Mechanics*
2016	Guest Lecturer, Stanford University, *Computational Cosmology and Astrophysics*
2014	Guest Lecturer, California Institute of Technology, *Cosmology and Galaxy Formation*
2009	Teaching Assistant, 1 quarter, Stanford University, *Black Holes*
2008	Course Grader, 1 quarter, Stanford University, *Introduction to Astrophysics*
2007	Teaching Assistant, 1 quarter, Stanford University, *Modern Physics Laboratory*
2007	Teaching Assistant, 1 quarter, Stanford University, *Physics*

MENTORING EXPERIENCES:

- Postdoctoral scholars : Dr. Keita Fukushima (10/2024 -),
  Dr. Boon Kiat Oh (10/2019 - 09/2022, → postdoc @ University of Connecticut)
- Ph. D. students : Eunwoo Chung (09/2024 -), Hyeonyong Kim (09/2023 -),
  Minyong Jung (09/2021 -), Seungjae Lee (09/2019 - 08/2025, → author/software engineer),
  Eun-jin Shin (03/2018 - 08/2023, → postdoc @ University of Cambridge),
  Yongseok Jo (03/2018 - 08/2022, → postdoc @ Columbia University & Flatiron Institute)
- M. Sci. students : Chan-mi Shin (09/2026 -), Yiyang Wang (09/2025 -), Hyeonu Seo (09/2025 -),
  Jun Yong Park (09/2022 - 02/2026), Ki-won Kim (09/2018 - 08/2019)
- B. Sci. interns : Minje Park (06/2026 - , from Sungkyunkwan U.), Hyungjun Kim (06/2026 - 08/2026),
  Minjun Choi (12/2025 - 07/2026), Seoyeon Park (12/2025 - , Thesis),
  Keonwoo Leo Kim (06/2025 - 08/2025, from McGill U.), Yiyang Wang (03/2025 - 07/2025),

Hyeonmin Eugene Lee (02/2025 - 08/2025, from UT Austin),
Daekyu Yoon (12/2024 -), Chaerin Jeong (08/2024 - 05/2025, from KyungHee U.),
Gyujin Kim (07/2024 - 04/2025), Geon Hee Kim (06/2024 - 02/2025),
Abdelrahman Helal (09/2023 - 12/2023, from Minerva U.),
Youngjun Cho (08/2023 - 11/2023, from UNIST),
Junhwa Jang (03/2023 - 06/2024, 03/2026 -), Hwan Hee Chung (03/2023 - 03/2024),
Donghwan Hyun (12/2022 - 02/2024, Thesis), Subo Hwang (12/2022 - 02/2024, Thesis),
Eunwoo Chung (12/2022 - 02/2024, Thesis),
Han Noh (07/2022 - 08/2022, from Andong U.),
Yong-uk Cho (06/2022 - 08/2022, from KyungHee U.),
Seoyoung Kim (12/2021 - 05/2024), Jun-Young Lee (12/2021 - 05/2025, Thesis),
Ho-sung Lee (09/2021 - 01/2022), Junhee Myong (06/2021 - 05/2023),
Ahram Lee (03/2021 - 05/2022), Hyeongmo Kim (01/2021 - 09/2021),
Minju Kum (12/2020 - 02/2021), Songyoun Park (10/2020 - 04/2022),
Hyeonyong Kim (09/2020 - 02/2023), Lael Shin (09/2020 - 03/2021, Thesis),
Eunhee Ko (09/2020 - 02/2021, Thesis), Su-un Lee (08/2020 - 10/2021, Thesis),
Hongjun An (08/2020 - 02/2021,Thesis), Hyerin Cho (01/2020 - 08/2020, from GIST),
Sangmin Bae (12/2019 - 07/2020), Eonho Chang (10/2019 - 12/2019, from UC Santa Cruz),
Joohyun Lee (09/2019 - 08/2021, Thesis), Seung-o Ha (07/2019 - 08/2019, from Postech),
Goojin Kwon (07/2019 - 09/2019, 06/2020 - 05/2021, 09/2021 - 11/2021, from U. Cambridge),
Minyong Jung (03/2019 - 08/2020), Sangmoon Lee (03/2019 - 08/2019, Thesis)

COMPUTING EXPERIENCES:

- Administrator of a Intel Xeon SP Gold + GPU cluster with 904 cores and max 1 TB memory/node (2018 -)
- Program experience: Enzo, Gadget, GIZMO, yt, MUSIC, VisIt, PartiView, HEALpix, CMBFast, etc.
- Language experience: C, C++, Fortran, IDL, Python, Java Applet, Visual Basic, Pascal, Matlab, etc.

RESEARCH INTERESTS:

- Coordinator, *AGORA* High-resolution Galaxy Simulations Comparison Project (AGORAsimulations.org)
- Galaxy Formation and Evolution using High-resolution Adaptive Mesh Refinement Simulations
- Radiative/Mechanical/Thermal Feedback from Massive Black Holes and Star-forming Molecular Clouds
- Growth of Supermassive Black Holes and Triggered Star Formation via Gas Inflow or Mergers

TALKS AND SEMINARS (SELECTED):

- **12th KIAS Workshop on Cosmology and Structure Formation**, *hosted by Korea Institute for Advanced Study, TBD, 11/2026*

- **Physics Colloquium**, *Korea Institute for Advanced Study,* "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives from the Hydrodynamics+*N*-body Hybrid Simulations", *10/07/2026*

- **National University of Singapore - Seoul National University (NUS-SNU) AI for Science Symposium**, *Seoul National University,* "How Can Machine Learning and Simulations Help Us To Study Our Universe (And Astronomically Large Datasets)?", *09/21/2026*

- **1st Galaxy Formation and AGORA Workshop in Europe**, *hosted by Lund University,* "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives from the Hydrodynamics+*N*-body Hybrid Simulations", *05/26/2026*

- **Institute of Astronomy (IoA) Colloquium**, *University of Cambridge,* "Reproducibility in Astrophysical Simulations: The AGORA Initiative and a 14-Year Human Experiment (In Memory of Joel Primack)", *05/21/2026*

- **8th Chirality in the Universe Beyond the Electroweak Scale (CUBES) Workshop**, *hosted by Center for Underground Physics and Center for Theoretical Physics of the Universe, Institute for Basic Science, Korea,* "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives from the Hydrodynamics+*N*-body Hybrid Simulations", *04/25/2026*

- **BK21 Annual Forum**, *Seoul National University,* "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives in the Era of High-resolution Simulations", *12/11/2025*

- **4th SNUARC Astronomy Workshop**, *Seoul National University, hosted by SNU Astronomy Research Center,* "Computational Cosmology Research", *08/21/2025*

- **Santa Cruz Galaxy Formation Workshop+13th *AGORA* Workshop**, *UC Santa Cruz,* "Past, Present, and Future of the AGORA Project (and a 13-Year Human Experiment)", *08/08/2025*

- **IBS CTPU-CGA Summer Workshop 2025**, *hosted by Center for Theoretical Physics of the Universe (Cosmology, Gravity and Astroparticle Physics Group), Institute for Basic Science, Korea,* "Cosmic Mystery of Supermassive Black Holes: Perspectives in the Era of High-resolution Simulations", *07/31/2025*

- **International Workshop on Galaxy Formation+*AGORA* in Asia 2025**, *Nakanoshima Center, hosted by University of Osaka,* "Simulating Dark Matter-Deficient Galaxy Formation in A Mini-bullet Satellite-satellite Galaxy Collision + The Status of the AGORA Project", *05/28/2025*

- **Artificial Intelligence and High Energy Physics (AI+HEP) in East Asia**, *Center for Theoretical Physics of the Universe (Particle Theory and Cosmology Group), Institute for Basic Science, Korea,* "How Can Machine Learning and Simulations Help Us To Study Our Universe?", *02/24/2025*

- **11th KIAS Workshop on Cosmology and Structure Formation**, *hosted by Korea Institute for Advanced Study,* "Reproducibility In Astrophysical Simulations and the AGORA Initiative (and a 12-Year Human Experiment)", *11/01/2024*

- **Physics Colloquium**, *Hanyang University,* "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Numerical Simulations", *10/30/2024*

- **Statistics Seminar**, *Seoul National University,* "How Can Machine Learning Help Us When Studying Our Universe (And Astronomically Large Datasets)?", *10/11/2024*

- **Symmetry and Structure of the Universe Workshop 2024**, *Jeonbuk National University,* "New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations", *07/29/2024*

- **Physics Colloquium**, *Korea University,* "New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations", *04/02/2024*

- **Inaugural Symposium of the Institute for Data Innovation in Science**, *Seoul National University,* "How Can Machine Learning Help Us When Studying Our Universe?", *03/28/2024*

- **Survey Science Group Workshop 2024**, *hosted by Korea Institute for Advanced Study,* "AGORA High-resolution Galaxy Simulations Comparison Initiative: 12 Years After Its Conception (and a 12-Year Human Experiment)", *01/31/2024*

- **Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback**, *hosted by Institute for Advanced Study in Physics, Zhejiang University and Shanghai Astronomical Observatory, Chinese Academy of Sciences,* "How Including New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations", *10/11/2023*

- **Korea Supercomputing Conference 2023**, *hosted by Korea Institute of Science and Technology Information,* "How Can Machine Learning Help Us When Studying and Simulating the Universe?", *08/22/2023*

- **DARWIN+Dwarf Galaxy Workshop**, *Korea Astronomy and Space Science Institute*, "How Can Machine Learning Help Us When Studying and Simulating the Universe?", *08/16/2023*
- **Santa Cruz Galaxy Formation Workshop+11th *AGORA* Workshop**, *UC Santa Cruz*, "Overview of the AGORA Project" *(joint with Joel Primack and Santi Roca-Fabrega), 08/11/2023*
- **XV International Conference on Gravitation, Astrophysics and Cosmology**, *hosted by Asia Pacific Center for Theoretical Physics*, "How Including New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations", *07/06/2023*
- **Cosmology Workshop on the Crossroad of Astrophysics and Particle Physics**, *hosted by Korea Institute for Advanced Study*, "Studying Dark Matter Using Realistic Simulations", *06/29/2023*
- **Astronomy Colloquium**, *Chungnam National University*, "Taking on the Mysteries in the Universe in the Era of High-resolution Simulations", *06/07/2023*
- **10th KIAS Workshop on Cosmology and Structure Formation**, *Korea Institute for Advanced Study*, "How Including New Physics Changes Galactic Evolution In Simulations: Perspectives in the Era of High-resolution Simulations", *10/28/2022*
- **9th East Asian Numerical Astrophysics Meeting**, *Tenbusu Hall, Okinawa, hosted by Chiba University*, "How Feedback Affects Galactic Evolution: Perspectives in the Era of High-resolution Simulations", *09/26/2022*
- **Santa Cruz Galaxy Formation Workshop+10th *AGORA* Workshop**, *UC Santa Cruz*, "The AGORA Project To Compare Zoom-in Cosmological Simulations By The Leading Hydro Codes of A MW-mass Galaxy" *(joint with Joel Primack), 08/19/2022*
- **International Astronomical Union Symposium 373: Resolving the Rise and Fall of Star Formation in Galaxies (at IAU General Assembly XXXI)**, *Busan Exhibition and Convention Center, co-hosted by Korean Astronomical Society and Korea Astronomy and Space Science Institute*, "How Feedback Affects Stellar and Galactic Evolution: Perspectives in the Era of High-resolution Simulations", *08/10/2022*
- **Center for Theoretical Physics Colloquium**, *Seoul National University*, "Observational Astrophysics Primer To Prep For JWST Star Party", *07/20/2022*
- **1st SNURC Astronomy Workshop**, *Seoul National University Siheung Convention Center, hosted by SNU Astronomy Research Center*, "Computational Cosmology Research", *06/30/2022*
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro (III)", *04/09/2022*
- **Beyond the Standard Model Workshop 2022**, *Chung-Ang University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", *02/10/2022*
- **Physics Colloquium**, *Ajou University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", *12/08/2021*
- **Annual Forum 2021**, *Samsung Science & Technology Foundation*, "Towards The Unabridged Understandings of The Growth of Supermassive Black Holes", *11/22/2021*
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro (II)", *11/20/2021*
- **Astronomy Colloquium**, *Shanghai Jiao Tong University*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations", *10/20/2021*
- **Astrophysics Seminar**, *Institute of Particle and Cosmos Physics, Universidad Complutense de Madrid*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", *05/14/2021*
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro", *05/01/2021*

- **Physics Colloquium**, *Yonsei University,* "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", *04/28/2021*

- **7th Galaxy Evolution Workshop**, *Institute of Astronomy and Astrophysics, Academia Sinica, National Taiwan University,* "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", *02/03/2021*

- **College of Education Seminar**, *Chosun University,* "Exoplanets: From Mythology To Science", *12/23/2020*

- **Physics Colloquium**, *Chung-Ang University,* "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", *12/07/2020*

- **Theoretical Astrophysics Colloquium**, *University of Osaka,* "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", *10/21/2020*

- **Numerical Astrophysics Workshop at Korea Supercomputing Conference 2020**, *hosted by Korea Institute of Science and Technology Information,* "Numerical Galaxy Formation in the Era of High-resolution Simulations and Machine Learning", *09/24/2020*

- **Science and Policy Advanced Research Course (SPARC)**, *Seoul National University,* "Exoplanets: From Mythology To Science", *01/07/2020*

- **Galaxy Formation and Evolution Across Cosmic Time**, *Institute of Astronomy and Astrophysics, Academia Sinica, National Taiwan University,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *12/11/2019*

- **Astronomy Colloquium**, *Yonsei University,* "Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges", *12/03/2019*

- **Center for Theoretical Physics Colloquium**, *Seoul National University,* "How To Make Your Own Universe On A Computer: Success and Challenges", *11/08/2019*

- **XAIENCE: Crossing-over the AI and Science**, *Seoul National University,* "Machine Learning In Astrophysics & Cosmology: Estimating Galactic Baryonic Properties from Their Dark Matter", *11/07/2019*

- **Special Physics Colloquium**, *Seoul National University,* "Cosmology and Exoplanets: How Scientists Turned Mythology Into Physics", *10/30/2019*

- **Zurich Meets Seoul**, *co-hosted by University of Zurich and Seoul National University,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *10/01/2019*

- **Santa Cruz Galaxy Formation Workshop+8th *AGORA* Workshop**, *UC Santa Cruz,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *08/08/2019*, "AGORA Project Update: 7 Years After Conception" *(joint with Joel Primack & Santi Roca-Fabrega), 08/09/2019*

- **4th Korean-American Kavli Frontiers of Science Symposium**, *co-sponsored by the U.S. National Academy of Sciences (NAS) and the Korean Academy of Science and Technology (KAST),* "Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges", *06/19/2019*

- **Astronomy Colloquium**, *Kyungpook National University,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *06/04/2019*

- **95th Korean Physical Society Meeting**, *hosted by KPS,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *04/26/2019*

- **Extremely Big Eyes on the Early Universe**, *Kavli Institute for the Physics and Mathematics of the Universe, University of Tokyo,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *03/25/2019*

- **Physics Colloquium**, *Seoul National University,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *12/05/2018*

- **Physics and Astronomy Colloquium**, *Sejong University,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *11/21/2018*

- **8th KIAS Workshop on Cosmology and Structure Formation**, *Korea Institute for Advanced Study,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *11/06/2018*

- **99th Korean Astronomical Society Meeting**, *hosted by KAS,* "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", *10/10/2018*

- **Physics Seminar**, *Korea Institute for Advanced Study,* "Dark Matter: A Computational Astrophysicist's Perspective", *10/08/2018*

- **Particle Physics Korea Meeting**, *Seoul National University,* "Dark Matter: A Computational Astrophysicist's Perspective", *09/28/2018*

- **Santa Cruz Galaxy Formation Workshop+7th *AGORA* Workshop**, *UC Santa Cruz,* "Overview of the AGORA High-resolution Cosmological Galaxy Simulations Comparison Project: 6 Years After Conception" *(joint with Joel Primack)*, *08/10/2018*

- **KDESci Meeting**, *Korea Institute for Advanced Study,* "Insights from the AGORA High-resolution Galaxy Simulations Comparison", *07/09/2018*

- **Galaxies A to Z Workshop**, *hosted by Center for Galaxy Evolution Research, Yonsei University,* "Studying the Interaction of Supermassive Black Holes and Their Host Galaxies", *06/26/2018*

- **Astronomy Colloquium**, *Seoul National University,* "Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges", *05/24/2018*

- **Santa Cruz Galaxy Formation Workshop+6th AGORA Workshop**, *UC Santa Cruz,* "Reproducibility: An Insight from the AGORA High-resolution Galaxy Simulations Comparison", *08/11/2017*

- **Einstein Fellows Symposium 2016**, *Harvard-Smithsonian Center for Astrophysics,* "Reproducibility: An Insight from the AGORA High-resolution Galaxy Simulations Comparison", *10/18/2016*

- **Santa Cruz Galaxy Formation Workshop+5th AGORA Workshop**, *UC Santa Cruz,* "The AGORA High-resolution Galaxy Simulations Comparison. II: Isolated Disk Test - Kickoff Discussion", *08/12/2016*

- **Einstein Fellows Symposium 2015**, *Harvard-Smithsonian Center for Astrophysics,* "Upcoming New Era in Numerical Galaxy Formation: New Challenges and Possibilities", *10/27/2015*

- **Santa Cruz Galaxy Formation Workshop+4th AGORA Workshop**, *UC Santa Cruz,* "AGORA Initiative and Infrastructure: Where We Stand and Why We Are Here", *08/21/2015*

- **KIPAC Tea Talk**, *SLAC National Accelerator Laboratory,* "Challenges in Numerical Galaxy Formation and the AGORA Initiative", *07/24/2015*

- **Pasadena Astronomy Postdoc Symposium 2015**, *UCLA Lake Arrowhead Conference Center,* "Challenges in Numerical Galaxy Formation and the AGORA Initiative", *04/09/2015*

- **Astronomy Theory Postdoc Lunch**, *California Institute of Technology,* "AGORA High-resolution Galaxy Simulations Comparison Project", *01/22/2015*

- **Astronomy Colloquium**, *Seoul National University,* "Galaxy Formation Simulations in the High-resolution Era: Success and Challenge", *12/10/2014*

- **Einstein Fellows Symposium 2014**, *Harvard-Smithsonian Center for Astrophysics,* "Challenges in Numerical Galaxy Formation and the AGORA Initiative", *10/28/2014*

- **Santa Cruz Galaxy Formation Workshop+3rd *AGORA* Workshop**, *UC Santa Cruz,* "Status of the AGORA High-resolution Cosmological Galaxy Simulations Comparison Project: 2 Years After Conception" *(joint with Joel Primack)*, *08/15/2014*

- **Carnegie Observatories Colloquium Series**, *Carnegie Observatories*, "Rethinking Galaxy Simulations in the High-resolution Era", *03/18/2014*
- **Santa Cruz Galaxy Formation Workshop+2nd *AGORA* Workshop**, *UC Santa Cruz*, "The *AGORA* Project: Initial Conditions and the Proof-of-concept Test", *08/16/2013*
- **Cosmology Seminar**, *Stanford University*, "Rethinking Galaxy Simulations in the High-resolution Era", *05/20/2013*
- **Santa Cruz Galaxy Formation Workshop+Starting Workshop for the AGORA High-resolution Galaxy Simulations Comparison Project**, *UC Santa Cruz*, "Galaxy Formation with Radiating Molecular Cloud Particles", *08/17/2012*
- **Workshop for Korean Young Cosmologists**, *Korea Astronomy and Space Science Institute*, "High-resolution Galaxy Formation with Massive Black Holes and Radiating Star Clusters", *06/25/2012*
- **SLAC Association for Student Seminars**, *SLAC National Accelerator Laboratory*, *05/11/2011*
- **KIPAC Tea Talk**, *SLAC National Accelerator Laboratory*, "Galaxy Formation and Mergers with Self-consistently Modeled Stars and Massive Black Holes", *04/08/2011*
- **Friday Lunchtime Astrophysics Seminar**, *UC Santa Cruz*, "Towards An Unabridged Understanding of The Coevolution of Galaxies and Massive Black Holes: What Have Simulators Tried? Why So Hard?", *12/10/2010*
- **Computational Astrophysics Group Seminar**, *University of Chicago*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *12/03/2010*
- **Computational Cosmology and Galaxy Formation Seminar**, *Princeton University*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *11/15/2010*
- **Theoretical Astrophysics Center Seminar,** *UC Berkeley*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *10/25/2010*
- **Cosmoclub Seminar**, *UC Santa Cruz*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *10/18/2010*
- **Friday Astro Lunch**, *UC Santa Barbara*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *10/15/2010*
- **Astronomy Tea Talk**, *California Institute of Technology*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution", *10/11/2010*
- **LCA Group Seminar**, *UC San Diego*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes", *09/30/2010*
- **Santa Cruz Galaxy Formation Workshop**, *UC Santa Cruz*, "Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes", *08/17/2010*
- **HIPACC Summer School on Galaxy Formation**, *University of California High-Performance AstroComputing Center*, "Galaxy Formation using Enzo with Properly Modeled Stars and Massive Black Holes", *07/28/2010*